

Sydney Housing Price Prediction and Decision Support System

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Assessment - 8.1 Distinction Task — Machine Learning Mini Project

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1. Introduction and Problem Definition

This project develops a machine learning system to predict residential property sale prices in Sydney. The aim is to investigate whether property characteristics can be used to estimate sale prices and provide a simple decision-support tool.

Predicting house prices is useful because property values can vary significantly depending on location and property characteristics. A machine learning model can identify patterns within historical sales and provide an estimated price for a new property. However, the system is intended as a decision-support tool rather than a replacement for professional valuation.

Three suburbs were selected: Manly, Parramatta and Liverpool. They were chosen because they represent substantially different housing markets. The collected data showed clear differences, with median prices of approximately AUD 1,962,500 in Manly, AUD 650,000 in Parramatta and AUD 550,000 in Liverpool.

A dataset of 100 sold properties was manually collected from publicly available property information, with 34 properties from Manly, 33 from Parramatta and 33 from Liverpool. Features included suburb, property type, bedrooms, bathrooms, parking, listed area and sale date.

The main data-quality issue was missing information. Some listings did not provide features such as parking or area. The manually collected sample may also contain selection bias and inconsistencies. In addition, only three suburbs and 100 properties were included, so the dataset does not represent the complete Sydney housing market. Important factors such as renovation quality, property condition, views and exact micro-location were also unavailable.

	property_id	property_address	suburb	postcode	property_type	sale_price_aud	bedrooms	bathrooms	parking	listed_area_m2	...	missing_
0	M001	3/66 Osborne Road, Manly, NSW 2095	Manly	2095	Apartment	1650000	2	1.0	1.0	87.0	...	
1	M002	4/3 Cliff Street, Manly, NSW 2095	Manly	2095	Apartment	840000	1	1.0	NaN	61.0	...	
2	M003	2/115-117 Sydney Road, Manly, NSW 2095	Manly	2095	Unit	2125550	2	1.0	1.0	183.0	...	
3	M004	35 Smith Street, Manly, NSW 2095	Manly	2095	House	3500000	3	2.0	NaN	NaN	...	
4	M005	5/34 Pacific Street, Manly, NSW 2095	Manly	2095	Apartment	1350000	2	1.0	NaN	NaN	...	

5 rows × 23 columns

Dataset shape: (100, 23)

Column names:

`['property_id', 'property_address', 'suburb', 'postcode', 'property_type', 'sale_price_aud', 'bedrooms', 'bathrooms', 'parking', 'listed_area_m2', 'sale_date', 'sale_year', 'sale_month', 'missing_feature_count', 'source_page', 'source_url', 'Unnamed: 16', 'Unnamed: 17', 'Unnamed: 18', 'Unnamed: 19', 'Unnamed: 20', 'Unnamed: 21', 'verification_status']`

2. Data Understanding and Feature Engineering

Exploratory analysis showed substantial variation in sale prices.

Figure 1: Overall Sale Price Distribution

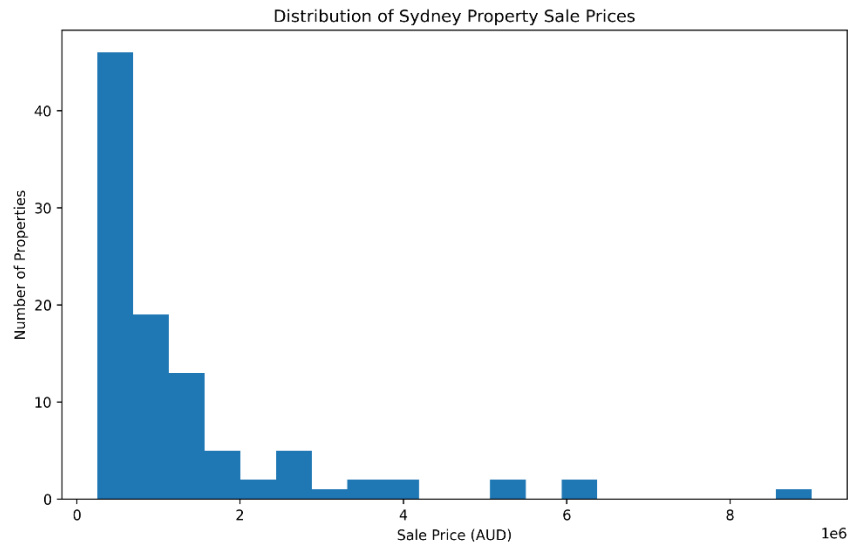
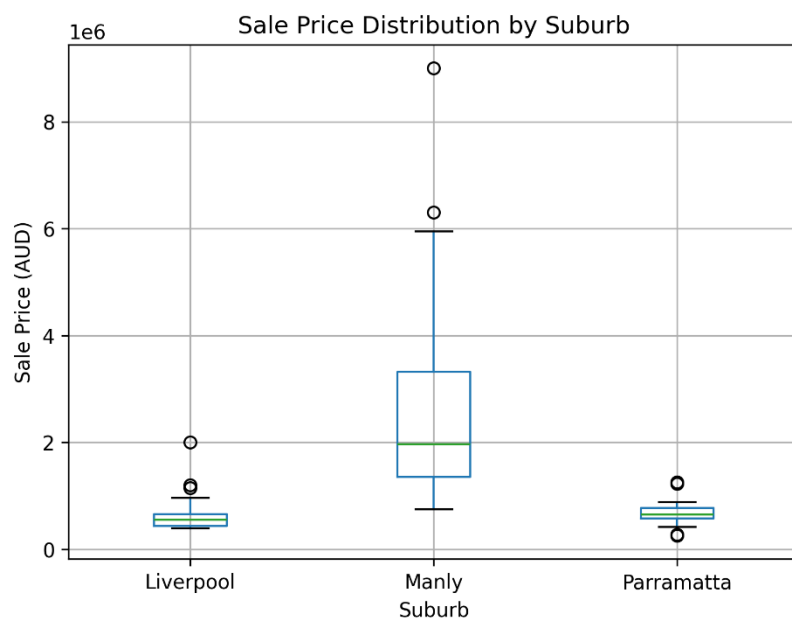


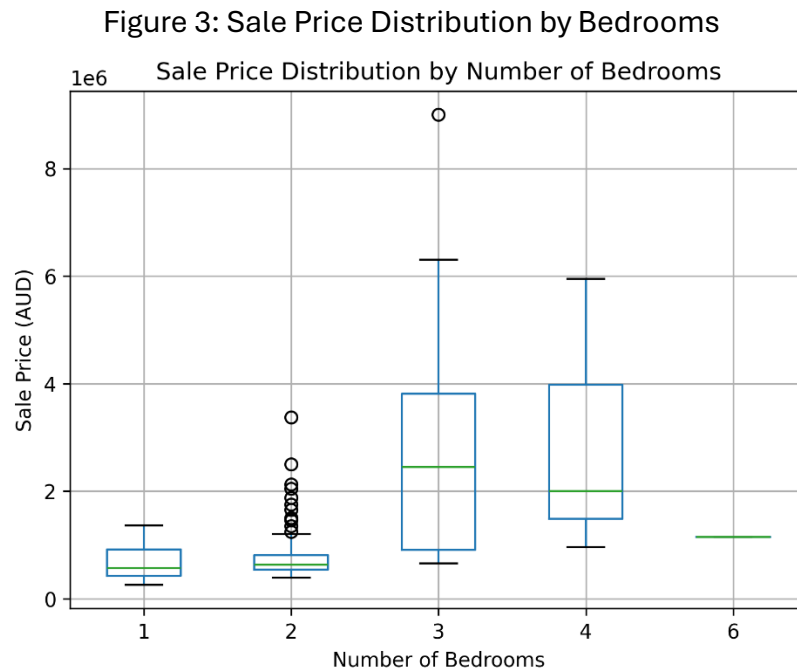
Figure 2: Sale Price Distribution by Suburb



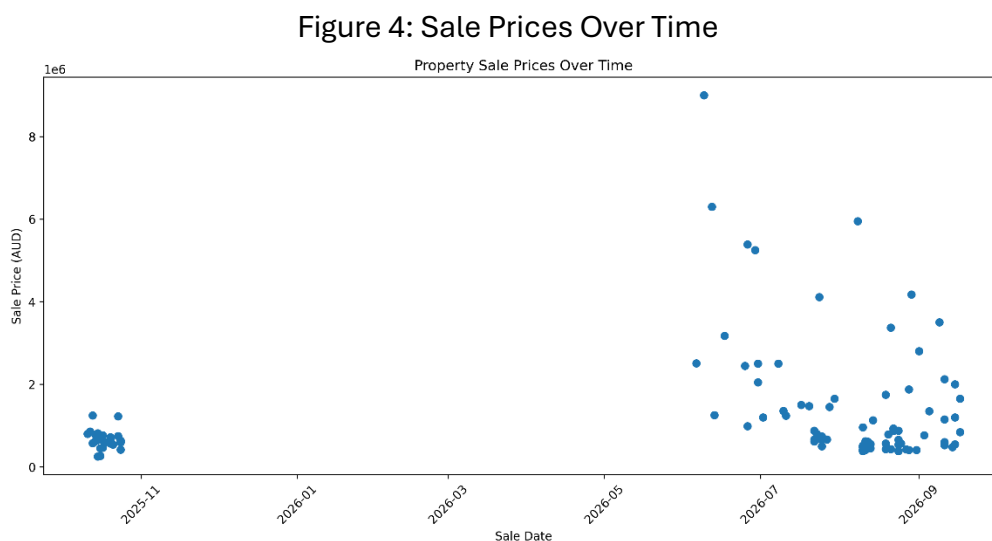
The suburb comparison showed that Manly generally had considerably higher prices than Parramatta and Liverpool. This supported the expectation that suburb would be one of the strongest predictors.

Property type also showed an important difference. Houses had a median sale price of approximately AUD 2,750,000, compared with AUD 668,500 for apartments, AUD 605,000 for units and AUD 262,500 for studios.

The number of bedrooms showed a moderate positive relationship with sale price, with a correlation of approximately $r = 0.439$.



Sale prices were also examined over time.



Three variables were initially expected to have the strongest influence: suburb, property type and bedrooms. These expectations were based on the differences observed during exploratory analysis.

Feature engineering added useful temporal and availability information, including sale year, sale month, sale quarter and an area-availability indicator. Categorical variables were also transformed through the preprocessing pipeline so that they could be used by the regression models.

3. Model Development and Evaluation

Three regression models were selected to represent different approaches:

Linear Regression — used as a simple baseline.

Random Forest Regression — selected because it can capture nonlinear relationships and interactions.

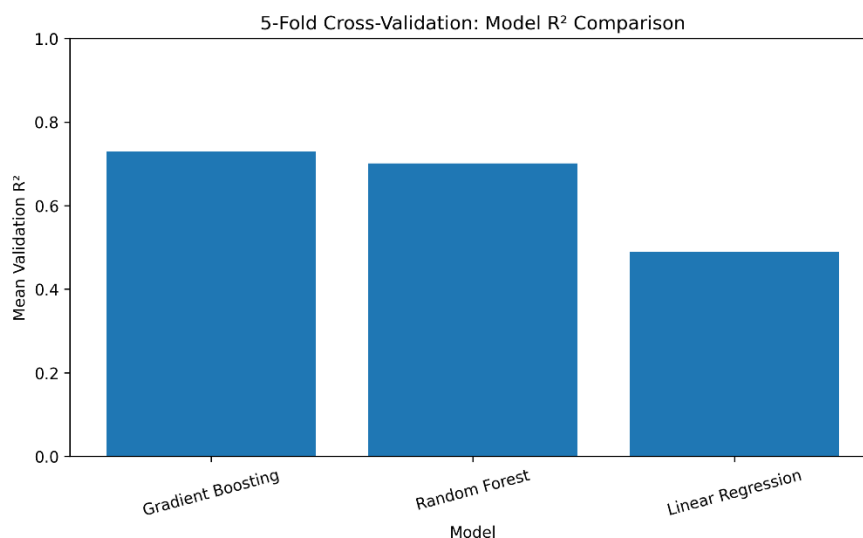
Gradient Boosting Regression — selected because it builds trees sequentially to reduce prediction errors.

Before training, the expectation was that the tree-based models would outperform Linear Regression because property prices are unlikely to have purely linear relationships. Gradient Boosting was expected to be particularly competitive.

The models were evaluated using five-fold cross-validation and a separate 20% test set.

Model	CV R^2	Test R^2
<i>Linear Regression</i>	0.4904	0.5373
<i>Random Forest</i>	0.7012	0.7763
<i>Gradient Boosting</i>	0.7297	0.7816

Figure 5: Five-Fold Cross-Validation R^2 Comparison



The results supported the original expectation. Both tree-based models performed substantially better than Linear Regression. Gradient Boosting achieved the highest cross-validation R^2 and test R^2 . Its test RMSE was approximately AUD 676,554, while its cross-validation RMSE was approximately AUD 709,919.

The relatively similar cross-validation and test performance suggests that Gradient Boosting generalised reasonably well rather than showing severe overfitting. However, the dataset is small, so performance could change with a different sample.

Gradient Boosting was therefore selected as the final model because it provided the strongest overall combination of validation and test performance.

4. Prediction Failures

The five properties with the largest absolute prediction errors were M028, M031, M032, L029 and L007.

	property_id	property_address	suburb	postcode	property_type	sale_price_aud	bedrooms	bathrooms	parking	1
27	M028	4/51C Kangaroo Street, Manly, NSW 2095	Manly	2095	Unit	990000	1	1.0	1.0	
30	M031	12/14 Malvern Avenue, Manly, NSW 2095	Manly	2095	Apartment	1253000	1	1.0	1.0	
31	M032	5/19a Addison Road, Manly, NSW 2095	Manly	2095	Townhouse	6300000	3	2.0	2.0	
95	L029	1/10 Tobruk Avenue, Liverpool, NSW 2170	Liverpool	2170	Townhouse	962000	4	3.0	1.0	
73	L007	505/17 Shepherd Street, Liverpool, NSW 2170	Liverpool	2170	Unit	570000	2	2.0	1.0	

The largest error occurred for M028, where the actual sale price was approximately AUD 990,000 while the model predicted approximately AUD 2.97 million. The property had missing listed-area information, and its actual price was much lower than the model expected for the available characteristics.

M031 was also substantially overpredicted, with an actual price of approximately AUD 1.253 million and a prediction of approximately AUD 2.73 million.

```
=====
=====
Property: M028 - 4/51C Kangaroo Street, Manly, NSW 2095
Suburb: Manly
Actual sale price: $990,000

Suburb price statistics:
Median: $1,962,500
Mean: $2,629,869
=====
=====
Property: M031 - 12/14 Malvern Avenue, Manly, NSW 2095
Suburb: Manly
Actual sale price: $1,253,000

Suburb price statistics:
Median: $1,962,500
Mean: $2,629,869
=====
=====
Property: M032 - 5/19a Addison Road, Manly, NSW 2095
Suburb: Manly
Actual sale price: $6,300,000
```

Suburb price statistics:

Median: \$1,962,500

Mean: \$2,629,869

=====

Property: L029 - 1/10 Tobruk Avenue, Liverpool, NSW 2170

Suburb: Liverpool

Actual sale price: \$962,000

Suburb price statistics:

Median: \$550,000

Mean: \$646,273

=====

Property: L007 - 505/17 Shepherd Street, Liverpool, NSW 2170

Suburb: Liverpool

Actual sale price: \$570,000

Suburb price statistics:

Median: \$550,000

Mean: \$646,273

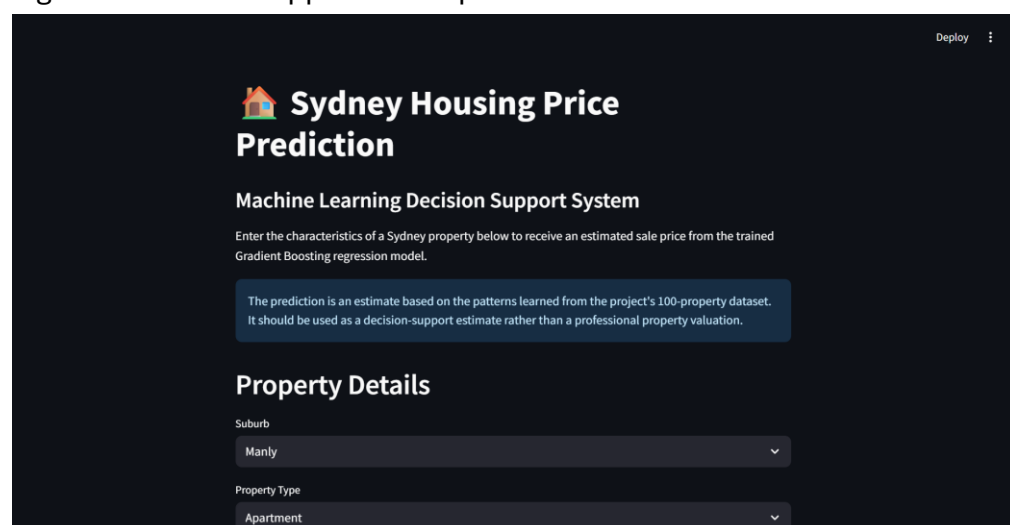
These cases demonstrate that structured features cannot capture every factor affecting an individual property. Missing information about renovation quality, property condition, views, exact location, land characteristics and interior features may explain some of the errors.

Predictions should therefore be treated cautiously when important information is missing or when a property has unusual characteristics that are poorly represented in the training data.

5. Deployment and Reflection

The final Gradient Boosting model was deployed using Streamlit. The application allows users to enter suburb, property type, bedrooms, bathrooms, parking, listed area and sale-date information and then receive an estimated sale price.

Figure 6: Streamlit Application Input



The screenshot shows a Streamlit application titled "Sydney Housing Price Prediction". It features a dark theme with a light blue header bar. The main content area is white. At the top, there's a house icon and the title "Sydney Housing Price Prediction". Below that, it says "Machine Learning Decision Support System". A text box instructs users to "Enter the characteristics of a Sydney property below to receive an estimated sale price from the trained Gradient Boosting regression model." A blue warning box states: "The prediction is an estimate based on the patterns learned from the project's 100-property dataset. It should be used as a decision-support estimate rather than a professional property valuation." Under the heading "Property Details", there are two dropdown menus. The first is labeled "Suburb" and has "Manly" selected. The second is labeled "Property Type" and has "Apartment" selected. In the top right corner, there is a "Deploy" button and a menu icon.

Property Details

Suburb

Manly

Property Type

Apartment

Bedrooms

2

Bathrooms

1

Parking Spaces

1

☒ Listed area information is available

Listed Area (m²)

100.00

Sale Information

Sale Year

2026

Sale Month

9

 Predict Sale Price

Figure 7: Streamlit Prediction Result

localhost:8501

Deploy

100.00

Sale Information

Sale Year

2026

Sale Month

9

 Predict Sale Price

Estimated Sale Price: AUD \$1,544,374

This prediction is generated by the project's trained Gradient Boosting regression pipeline.

The application can be run using:

```
pip install -r requirements.txt
```

```
streamlit run app/app.py
```

```
diduf@Didu_LENOVO MINGW64 /e/Deakin/Y2 S1 T2/SIT307 - Machine Learning/OnTrack/8
.ID/Sydney_Housing_ML_Project (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

diduf@Didu_LENOVO MINGW64 /e/Deakin/Y2 S1 T2/SIT307 - Machine Learning/OnTrack/8
.ID/Sydney_Housing_ML_Project (main)
$ python -m streamlit run app/app.py

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://172.20.10.5:8501
```

The project showed that data quality matters as much as model selection. Manual collection was time-consuming and missing information made modelling more difficult. The comparison of models showed that nonlinear ensemble methods were better suited to this dataset than a simple linear model.

There are also important fairness and bias considerations. The dataset contains only 100 properties from three suburbs, meaning the model may not perform equally well for properties or locations outside the training data. It should therefore not be treated as a professional valuation system.

With more time and resources, the project could be improved by collecting more properties and suburbs, adding land size and geographic information, including distances to transport and amenities, incorporating property condition and renovation information, and using natural language processing on real-estate descriptions. More extensive hyperparameter optimisation and testing on a larger independent dataset could also improve reliability.

Conclusion

The project successfully developed and evaluated three regression approaches and deployed the selected model as a Streamlit decision-support application. Gradient Boosting achieved the strongest results, with a cross-validation R^2 of 0.7297 and test R^2 of 0.7816. The prediction-error analysis also demonstrated the limitations of relying only on structured property information. Overall, the project shows how machine learning can support housing-price estimation while highlighting the importance of data quality, model evaluation and cautious interpretation.

References

Realestate.com.au — publicly available sold-property information used during dataset collection.

Scikit-learn documentation — regression models and evaluation methods.

Streamlit documentation — web application development.

Python documentation — programming language and supporting libraries.